Mumbai Education Trust's

INSTITUTE OF ENGINEERING, NASHIK.

COMPUTER ENGINEERING DEPARTMENT

Subject : DS **ASSIGNMENT NO – 01** **Unit : I**

1. **Define following term with an example :**
   (i) Data Structure (ii) ADT (iii) Algorithm (iv) Flowchart (v) Pseudo code
2. Give complete **classification of data structures** with one example of each.
3. Define **algorithms** and Explain its characteristics.
4. Define and Explain **ADT(Abstract Data Type).**
5. **Differentiate between: [5]**
   i) Static and Dynamic Data structures
   ii) Linear and Non-Linear data structure
   iii) Persistent and Ephemeral Data Structures
6. Explain the **Asymptotic Notation** Big O, Omega and Theta with suitable example.
   OR What is **Complexity Analysis of an Algorithm** ? Explain the **notations** used in the complexity analysis.
7. State Analysis of programming constructs - **Linear, Quadratic, Cubic, Logarithmic.**
   Or Discuss any **six classes of time complexity** with sample code
8. Explain the **Divide and Conquer Strategy** with suitable example. Comment on its time complexity.
9. Explain the **Greedy strategy** with suitable example. Comment on its time complexity.
10. What is **frequency count?** Why is frequency count important in the analysis of algorithm. Or Explain **step count method** with the help of suitable example
11. Write an **algorithm** to compute the sum of the digits of the given number. Justify that your algorithm satisfies all the characteristics of an algorithm.
12. Draw **flowchart** to check whether a given number is a perfect square of an integer. What is the time complexity of your algorithm.
13. Write **pseudo code** to find factorial of a number. Draw the **flowchart** for the same.

********** THE END **********

Study material provided by: Vishwajeet Londhe

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